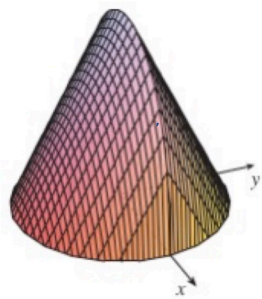


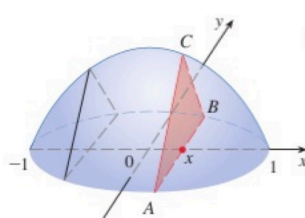
Finding Volume using cross-sectional Areas:

Example:

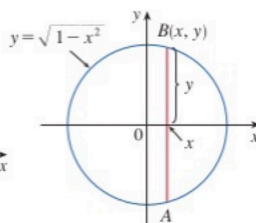
this picture shows a solid with a circular base of radius 1 parallel cross-sections perpendicular to the base are equilateral triangles. Find the volume of the solid.



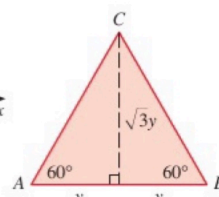
Sln: Let's take the circle $x^2 + y^2 = 1$.



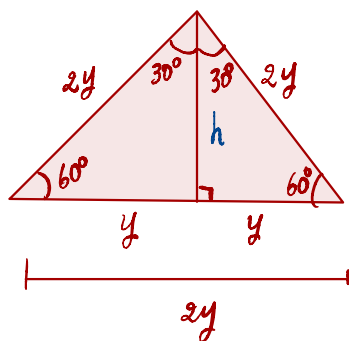
(a) The solid



(b) Its base



(c) A cross-section



$$(2y)^2 = y^2 + h^2$$

$$4y^2 - y^2 = h^2$$

$$3y^2 = h^2$$

$$h = \sqrt{3}y$$

So,

$$A(x) = \frac{1}{2} (2y) (\sqrt{3}y) = \sqrt{3}y^2.$$

Since $y^2 = 1 - x^2$ then

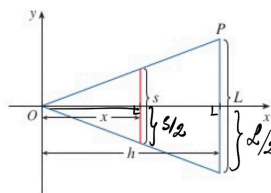
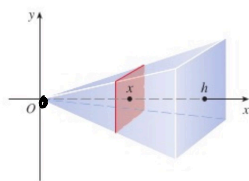
$$A(x) = \sqrt{3} (1 - x^2)$$

$$\text{So, } V = \int_{-1}^1 A(x) dx = \int_{-1}^1 \sqrt{3} (1 - x^2) dx = 2\sqrt{3} \int_0^1 (1 - x^2) dx.$$

$$= 2\sqrt{3} \left(x - \frac{x^3}{3} \right) \Big|_0^1 = 2\sqrt{3} \left(1 - \frac{1}{3} \right) = \frac{4\sqrt{3}}{3}.$$

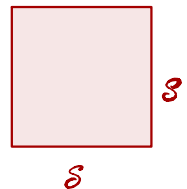
Example. Find the volume of a pyramid whose base is a square with side L and whose height is h .

Sln.



$$\frac{x}{h} = \frac{3/2}{L/2} = \frac{3}{L} \quad \text{then}$$

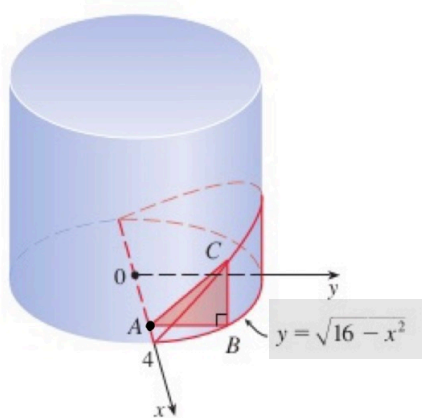
$$g = \frac{L \cdot x}{h}$$



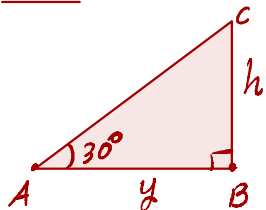
$$A(x) = g^2 = \left(\frac{Lx}{h}\right)^2 = \frac{L^2}{h^2} \cdot x^2. \quad \text{then}$$

$$\begin{aligned} V &= \int_0^h A(x) dx = \int_0^h \frac{L^2}{h^2} x^2 dx = \frac{L^2}{h^2} \int_0^h x^2 dx \\ &= \frac{L^2}{h^2} \left\{ \frac{x^3}{3} \Big|_0^h \right\} = \frac{L^2 h^3}{3 h^2} = \frac{1}{3} L^2 h. \end{aligned}$$

Example. A wedge is cut out of a circular cylinder of radius 4 by two planes. One plane is perpendicular to the axis of the cylinder the other plane intersects the first at an angle of 30° along the diameter of cylinder. Find the volume of the wedge.



soln. Notice that this triangle:



$$A(x) = \frac{1}{2} y \cdot h$$

$$A(x) = \frac{1}{2} \sqrt{16 - x^2} \cdot h$$

$$\tan 30^\circ = \frac{h}{y} \Rightarrow h = y \cdot \tan 30^\circ = \frac{y}{\sqrt{3}} = \frac{\sqrt{16 - x^2}}{\sqrt{3}}$$

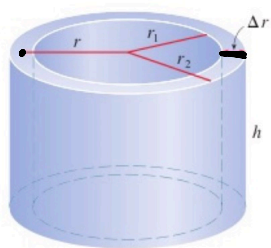
$$A(x) = \frac{1}{2} \sqrt{16 - x^2} \cdot \frac{\sqrt{16 - x^2}}{\sqrt{3}}$$

$$A(x) = \frac{16 - x^2}{2\sqrt{3}}$$

$$V = \int_{-4}^4 \frac{16 - x^2}{2\sqrt{3}} dx = \frac{2}{2\sqrt{3}} \int_0^4 (16 - x^2) dx = \frac{1}{\sqrt{3}} \left\{ 16x - \frac{x^3}{3} \Big|_0^4 \right\} = \frac{128}{3\sqrt{3}}$$

5.3 Volumes by cylindrical shells.

* The method of cylindrical shells.



Notice that the volume of this solid.

$$V = V_2 - V_1$$

$$V = \pi r_2^2 \cdot h - \pi r_1^2 \cdot h$$

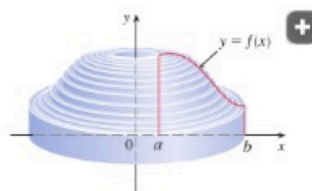
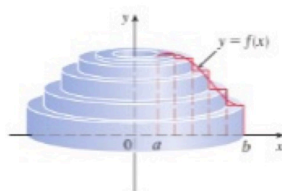
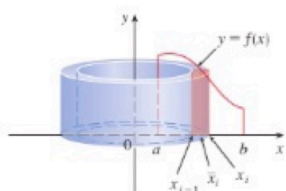
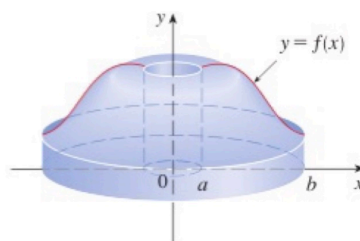
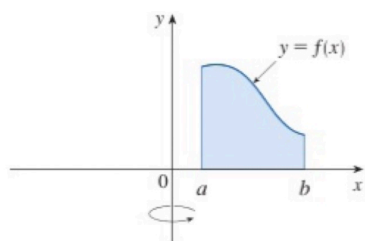
$$V = \pi (r_2^2 - r_1^2) \cdot h$$

$$V = 2\pi \left(\frac{r_2 + r_1}{2} \right) (r_2 - r_1) \cdot h$$

Notice that $\Delta r = r_2 - r_1$ and $r = \frac{r_1 + r_2}{2}$ so.

$$V = 2\pi r h \Delta r$$

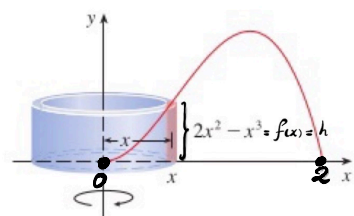
Now, let S be the solid obtained by rotating about y -axis the region bounded by $y = f(x)$ $\{f(x) \geq 0\}$ $y = 0$, $x = a$, $x = b$.



Def. The volume of the solid above, obtained by rotating about the y -axis the region under the curve $y = f(x)$ from a to b is.

$$V = \int_a^b 2\pi x f(x) dx$$

Example. Find the volume of the solid obtained by rotating about the y -axis the region bounded by $y = 2x^2 - x^3$ and $y = 0$.



Sln: Solving

$$0 = 2x^2 - x^3$$

$$0 = x^2(x - 2) \Rightarrow x = 0, x = 2$$

Using the cylinders. we have

$$V = \int_0^2 2\pi x f(x) dx = 2\pi \int_0^2 x(2x^2 - x^3) dx$$

$$V = 2\pi \int_0^2 (2x^3 - x^4) dx$$

$$V = 2\pi \left\{ \frac{x^4}{2} - \frac{x^5}{5} \right\}_0^2 = \frac{16}{5} \pi.$$

For you! try solve this problem using cross-sections perpendicular to the y-axis.